YUNHAI HAN

(858)214-4416 ⋄ yhan389@gatech.edu ⋄ https://y8han.github.io

EDUCATION BACKGROUND

Georgia Institute of Technology 05/2022 - present
Ph.D. in Robotics

Advisor: Harish Ravichandar

Advanced to Ph.D. candidacy in April 2026. Committee: Zsolt Kira, Danfei Xu, Yunzhu Li, and Dinesh Jayaraman.

University of California, San Diego (UCSD) 09/2019 - 06/2021
M.S. in Mechanical Engineering *GPA: 3.846/4.00*

· **Relevant Course:** Robotics

Yanshan University 09/2015 - 07/2019
B.S. in Mechanical Engineering *GPA: 3.761/4.5, Major GPA: 3.804/4.5*

· **Relevant Course:** Mechatronics *Ranking:* 2^{nd} of 594 (First six semesters)

RESEARCH INTERESTS

Structured robot learning, learning dexterous skills from human videos, robot behavior models, tactile adaptation, dynamical systems for manipulation

PUBLICATIONS

- **Han. Y***, Qiu. J*, Bai. L, Xiao. Z, Zeng. Z, Liu. Y, Yang. Z, Jain. S, Ma. W, Fu. J, Zheng. Y, Natarajan. M, Irshad. M. Z, Shaw. K, Gombolay. M, Kira. Z, Ravichandar. H, "Video2Sim2Real: Full-Stack Autonomous Dexterous Skill Acquisition from a Single Human Video" (* Equal Contribution), Under Review
- Yu. K*, Zhang. H*, Ravichandar. H, **Han. Y**, Gao. R, "OmniTacTune: Policy-Agnostic Real-World RL for Tactile Residual Adaptation of Visual Policies" (* Equal Contribution), Best Paper Award at RSS Tac for FM Workshop 2026
- **Han. Y**, Fu. J*, Bai. L*, Xiao. Z*, Yang. Z*, Choudhary. Y, Jha. K, Kong. C, Kousik. S, Ravichandar. H, "Going with the Flow: Koopman Behavioral Models as Pseudo Planners for Visuo-Motor Dexterity" (* Equal Contribution among co-second authors), Under Review
- Kim. J, **Han. Y**, Ravichandar. H, Ha. S, "Safe Navigation of Bipedal Robots via Koopman Operator-Based Model Predictive Control", IROS 2026
- Liu. Y*, Shin. W*, **Han. Y**, Chen. Z, Ravichandar. H, Xu. D, "ImMimic: Cross-Domain Imitation from Human Videos via Mapping and Interpolation" (* Equal Contribution), CoRL 2025 Oral; RSS Dex Workshop Spotlight
- Guo. H*, **Han. Y***, Ravichandar. H, "On the Surprising Effectiveness of Spectral Clipping in Learning Stable Linear and Latent-Linear Dynamical Systems" (* Equal Contribution; Guo is my master student advisee), Under Review

- Yang. Z, **Han. Y**, Ravichandar. H, "AsymDex: Leveraging Asymmetry and Relative Motion in Learning Bimanual Dexterity", Spotlight at RSS Whole-body Control and Bimanual Manipulation Workshop 2025; Poster at CoRL Workshop 2024
- Chen. H, Abuduweili. A*, Agrawal. A*, **Han. Y***, Ravichandar. H, Liu. C, Ichnowski. J, "KOROL: Learning Visualizable Object Feature with Koopman Operator Rollout for Manipulation" (* Equal Contribution), CoRL 2024
- **Han. Y**, Chen. Z, Williams. K. A, Ravichandar. H, "Learning Prehensile Dexterity by Imitating and Emulating State-only Observations", IEEE Robotics and Automation Letters
- Yu. K*, **Han. Y***, Wang. Q, Saxena. V, Xu. D, Zhao. Y, "MimicTouch: Leveraging Multi-modal Human Tactile Demonstrations for Contact-rich Manipulation" (* Equal Contribution; Yu is my master student advisee), CoRL 2024; Best Paper Award at NeurIPS Touch Processing Workshop
- **Han. Y**, Xie. M, Zhao. Y, Ravichandar. H, "On the Utility of Koopman Operator Theory in Learning Dexterous Manipulation Skills", CoRL 2023 Oral
- M. E. Cao, J. Warnke, **Han. Y**, Ni. X, Zhao. Y, Coogan. S, "Leveraging Heterogeneous Capabilities in Multi-Agent Systems for Environmental Conflict Resolution", SSRR 2022
- **Han. Y** and Martinez. S, "A Numerical Verification Framework for Differential Privacy in Estimation", L-CSS & ACC 2022
- Christensen. H. I, Paz. D, Zhang. H, Meyer. D, Xiang. H, **Han. Y**, Liu. Y, Liang. A, Zhong. Z, Tang. S, "Autonomous Vehicles for Micro-Mobility", Autonomous Intelligent Systems 2021
- **Han. Y***, Yu. K*, Batra. R, Boyd. N, Mehta. C, Zhao. T, She. Y, Hutchinson. S, Zhao. Y, "Learning Generalizable Vision-Tactile Robotic Grasping Strategy for Deformable Objects via Transformer", IEEE/ASME Transactions on Mechatronics
- Liu. F*, Li. Z*, **Han. Y**, Lu. J, Richter. F, Yip. M. C, "*Real-to-Sim* Registration of Deformable Soft Tissue with Position-Based Dynamics for Surgical Robot Autonomy", ICRA 2021
- **Han. Y***, Liu. Y*, Paz. D, Christensen. H. I, "Auto-calibration Method Using Stop Signs for Urban Autonomous Driving Applications", ICRA 2021
- **Han. Y**, Liu. F, Yip. M. C, "A 2D Surgical Simulation Framework for Tool-Tissue Interaction", IROS 2020 Workshop Spotlight

HONORS AND AWARDS

Qualcomm Innovation Fellowship Finalist 2026
Best Paper Award, RSS Tac for FM Workshop 2026
Best Paper Award, NeurIPS Touch Processing Workshop 2024
Georgia Tech IRIM Robotics PhD Fellowship
National Scholarship from Chinese Ministry of Education

PROFESSIONAL SERVICE

Workshop Co-organizer – Bimanual Dexterous Tool Use Workshop at CoRL 2026
GT RoboGrads Executive Board – Research Vice President
Conference Session Chair – ACC
Conference Reviewer – ICRA, IROS, RSS, CoRL, NeurIPS, ICLR, AIM, SSRR
Journal Reviewer – RA-L, IEEE T-RL, OJ-CSYS

TEACHING EXPERIENCE

MAE145: Robotic Planning & Estimation, UCSD Winter 2021
Teaching Assistant
MAE146: Introduction to ML Algorithms, UCSD Spring 2021
Teaching Assistant

RESEARCH AND WORKING EXPERIENCE

NVIDIA Seattle Robotics Lab 05/2026 - Present
Research Intern – Humanoid robot learning from offline human manipulation videos.
Georgia Institute of Technology 05/2022 - Present
Graduate Research Assistant – Structured robot learning for dexterous manipulation from human videos, tactile feedback, and behavioral dynamical systems.
Georgia Institute of Technology 06/2021 - 05/2022
Research Assistant – Robot learning and manipulation research before the Ph.D. program.

TECHNICAL SKILLS

Programming	C/C++, Python, MATLAB/Simulink
Tool	PyTorch, STM32, ROS, Drake, Git, Linux, LaTeX
Language	Proficient in English and Chinese